ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

DYNAMICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 2nd year final examination: kinetics of particles and systems of particles (Newton's second law, work-energy, momentum and angular momentum, steady mass flow), plane kinematics of rigid bodies, moments of inertia, plane kinetics of rigid bodies, and vibration.

Written by Academic Central for practice, following engineering dynamics modules at South African universities (e.g. University of Pretoria MSD 210 Dynamics: kinetics of systems of particles, plane kinematics and kinetics of rigid bodies, vibration). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Draw free-body diagrams where they help. Show ALL your working, including formulae and substitutions.
3. Use g = 9.81 m/s². Give answers correct to TWO decimal places, with SI units.
4. Moments of inertia: slender rod about its centre mL²/12; solid disc or cylinder about its axis ½mr².
5. Number your answers exactly as the questions are numbered.

QUESTION 1: KINETICS OF PARTICLES AND SYSTEMS OF PARTICLES

1.1 A 20 kg block on a horizontal table (μk = 0.25) is connected by a light string over a frictionless pulley to a 10 kg mass that hangs freely. The system is released from rest. Calculate the acceleration and the tension in the string. (6)

1.2 A 5 kg package slides from rest down a smooth chute, dropping a vertical height of 1.5 m, and then moves along a smooth horizontal surface into a buffer spring with k = 2 000 N/m. Calculate the maximum compression of the spring. (5)

1.3 A 60 kg person standing in a stationary 120 kg boat jumps horizontally off the back at 3 m/s. Ignore water resistance. Calculate the speed of the boat just afterwards. (3)

1.4 A 0.2 kg ball moves in a horizontal circle of radius 0.5 m at 2 m/s on a smooth table, held by a string through a hole in the centre. The string is pulled until the radius is 0.25 m. Calculate the new speed. (3)

1.5 A water jet with a flow rate of 0.02 m³/s and a speed of 15 m/s strikes a fixed flat plate at right angles and leaves along the plate. Calculate the force on the plate. (ρ = 1 000 kg/m³) (3)

[20]

QUESTION 2: PLANE KINEMATICS OF RIGID BODIES

FIGURE 1 shows a slider-crank mechanism. The crank OA = 100 mm turns clockwise at a constant 300 r/min. The connecting rod AB = 400 mm. B slides along the horizontal line through O. In the position shown, OA is vertical.

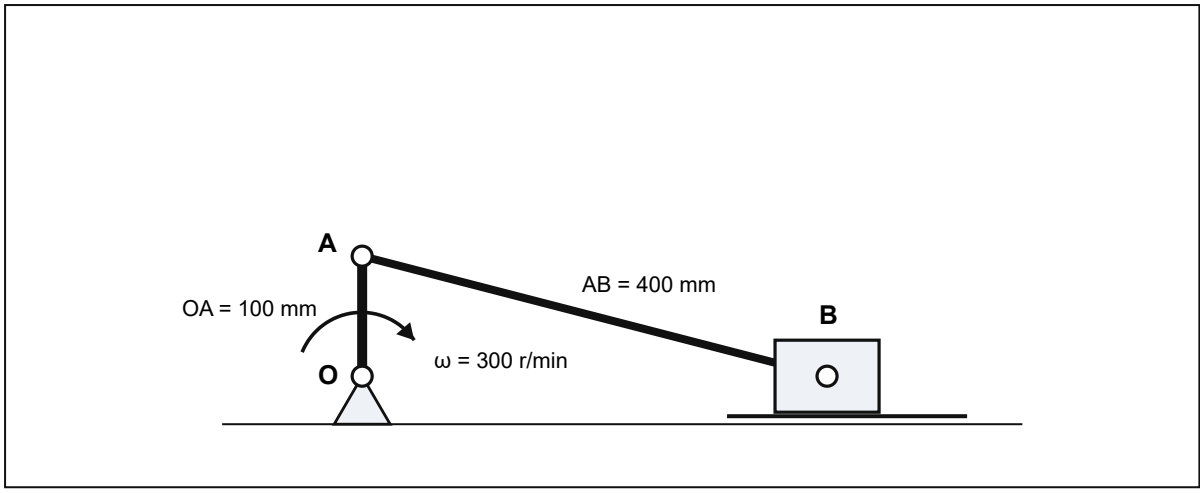


FIGURE 1: Slider-crank mechanism (OA vertical)

2.1 Calculate the angular velocity of the crank in rad/s and the velocity of A. (2)

2.2 Show that the angular velocity of AB is zero in this position, and find the velocity of the slider B. (4)

2.3 Calculate the acceleration of A. (2)

2.4 Calculate the angular acceleration of AB. (5)

2.5 Calculate the acceleration of the slider B and give its direction. (3)

2.6 A wheel of radius 0.3 m rolls without slipping, with its centre moving at 6 m/s. Calculate its angular velocity, and the velocities of the top point and of the contact point. (5)

2.7 Explain what the instantaneous centre of zero velocity is, and how it is found for a body whose velocities at two points are known. (4)

[25]

QUESTION 3: MOMENTS OF INERTIA

3.1 Calculate the mass moment of inertia of a 20 kg solid disc of radius 0.25 m about its axis. (2)

3.2 A slender rod has a mass of 3 kg and a length of 1.2 m.

3.2.1 Calculate its moment of inertia about one end. (2)

3.2.2 Use the parallel-axis theorem to confirm your answer to 3.2.1. (2)

3.3 A pendulum is made from the rod in 3.2, pinned at its top end O, with a 2 kg solid disc of radius 0.1 m fixed to the bottom end so that the centre of the disc is 1.3 m below O. Calculate the moment of inertia of the pendulum about O. (5)

3.4 Calculate the radius of gyration of the disc in 3.1, and explain what the radius of gyration means. (4)

[15]

QUESTION 4: PLANE KINETICS OF RIGID BODIES

4.1 The disc in 3.1 is driven by a motor torque of 10 N·m against a bearing friction torque of 2 N·m.

4.1.1 Calculate its angular acceleration. (3)

4.1.2 Calculate the time to reach 1 500 r/min from rest. (2)

4.2 The 3 kg, 1.2 m rod in 3.2 is pinned at one end and released from rest in a horizontal position.

4.2.1 Calculate its angular acceleration at the moment of release. (3)

4.2.2 Calculate the vertical reaction at the pin at the moment of release. (3)

4.2.3 Use the work-energy method to find its angular velocity when it swings through the vertical position. (4)

4.3 A solid cylinder rolls without slipping down a slope inclined at 20°, starting from rest.

4.3.1 Show that its acceleration is $(2/3)g \sin \theta$, and calculate it. (4)

4.3.2 Calculate the smallest coefficient of static friction needed to prevent slipping. (3)

4.3.3 Calculate its speed after it has rolled 5 m. (3)

[25]

QUESTION 5: VIBRATION

A 10 kg machine is mounted on springs with a total stiffness of 4 000 N/m and a damper with c = 80 N·s/m.

5.1 Calculate the undamped natural frequency in rad/s and in Hz. (3)

5.2 Calculate the critical damping coefficient, the damping ratio and the damped natural frequency. (4)

5.3 The machine is displaced and released. Calculate the logarithmic decrement. (3)

5.4 A harmonic force with an amplitude of 100 N acts on the machine.

5.4.1 Calculate the amplitude of vibration when the forcing frequency is 10 rad/s. (3)

5.4.2 Calculate the amplitude at resonance. (2)

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TOTAL: 100

END OF QUESTION PAPER